Zerui Chen | 陈泽瑞

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Graduating Jul 2027 · seeking research engineer / fellowship roles from summer 2027

Education

University of Cambridge | BA (Hons) Computer Science | *Cambridge, UK* **Expected Jul 2027**

- **Key Coursework:** Artificial Intelligence, Advanced Data Science, Bayesian Inference, Deep Neural Networks, Natural Language Processing, Geometric Deep Learning, Reinforcement Learning (expected by graduation)
- **Year 2 Grading: First Class** - course rank **18 / 129**

Hwa Chong Institution | GCE A-Levels: 4 H2 **AAAA/AA**, H3 **Distinction** | *Singapore* **Jan 2016 - Dec 2021**

Publications & Research

Search Pathology Diagnosis for Automated Discovery **Oct 2026**

Co-first author | ***NeurIPS 2026*** *workshops:* ***Oral****, MetaAgents ·* ***Poster****, Interpreting Agent Behavior (IAB)*

- Built a **system-agnostic** toolkit to detect **2 types of search pathologies** in automated discovery systems (ADS), based on a unified intermediate representation of traces and a bias-robust protocol to tune and evaluate the detectors.
- Demonstrated the detectors through a case study on a **SOTA ADS**, yielding **28% improvement in sampling efficiency**.

Evolution Fine-Tuning: Learning to Discover Across 371 Optimization Tasks **May 2026**

Co-author | ***NeurIPS 2026*** *(poster) ·* ***Oral****, AID-Wild Workshop @ ACM CAIS 2026 · arXiv:2606.29082*

- Contributed **nanodiscover** as the **test-time-RL scaffold** behind the paper's test-time-RL results.

nanodiscover **2026**

Co-author | *Open-source research software*

- A clean reimplementation of TTT-Discover, a **SOTA test-time-learning automated discovery system**, replacing its Tinker API dependency (≈$500 per problem) with local vLLM/DeepSpeed/Ray inference and training. Fixes upstream bugs, reproduces the published results up to sampling stochasticity, and was adopted by Evolution Fine-Tuning.

Comparative study of decision trees vs deep learning models for atrial tachycardia diagnosis **Jun 2022**

First author | *arXiv:2206.07515*

Professional Experience

WorldQuant | Quantitative Research Intern | *Singapore* **Jul 2026 - Sep 2026**

- Designed and built a **tiered forecast-evaluation suite** for residualized return forecasts: **headline metrics** spanning calibrated accuracy and a PnL proxy, **diagnostics** for calibration shape, tail effects and realisation dynamics, and **robust inference** throughout via HAC standard errors and simultaneous confidence bands, merged into production.
- Produced a one-parameter **training recipe** for **LightGBM** return-forecasting models: a single dial walks the **ICIR–Sharpe efficiency frontier** by tuning tree complexity and loss sensitivity to yield training hyperparameters.

ByteDance, Global E-Commerce | Machine Learning Engineer Intern | *Singapore* **Jul 2025 - Sep 2025**

- Researched and shipped **auxiliary objectives** enabling dense ranking supervision under very sparse positive feedback achieving **+0.3% revenue lift** in online A/B tests, and prototyped **model architectural** optimizations using recurrent transformers with per-token early exit for TikTok Shop's fine-ranking model, trained on in-house **HPC** clusters.
- Built internal tools for interactive model architecture **visualization** and automated **training behavior analysis**.

[Competition] NUS Medical Grand Challenge | Full-Stack & LLM Developer | *Global* **Feb 2024 - Aug 2024**

- Founded medical-AI startup *Tomy* for remote stoma care and emerged **Grand Champion** amongst 40 teams from universities across Asia, partnering with **5 Singapore public hospitals** with deployed clinical trials.
- Implemented and deployed image classification API using few-shot prompted vision language model, achieving **100% recall** and **80% precision** in 4-fold cross-validation with extremely scarce training data.

Agency for Science, Technology and Research | Bio-AI Research Intern | *Singapore* **Jun 2020 - May 2021**

- Developed a LSTM-CNN deep learning model for atrial tachycardia diagnosis, reaching SOTA-comparable 81% accuracy while being highly data-efficient; parallelized data ETL pipeline to achieve **100× throughput**.

Technical Skills

- **Machine Learning** with Python using PyTorch, vLLM, DeepSpeed, Ray, LightGBM for LLM inference and fine-tuning and deep learning for NLP, CV, RecSys, with Lightning, Slurm and AWS for experiment orchestration.
- **Data ETL & Analysis** with SQL, MongoDB, NumPy, Pandas, Matplotlib
- **General Development** with C++, Java, OCaml, Rust, TypeScript, Swift
- **Languages:** English (native), Mandarin Chinese (native).